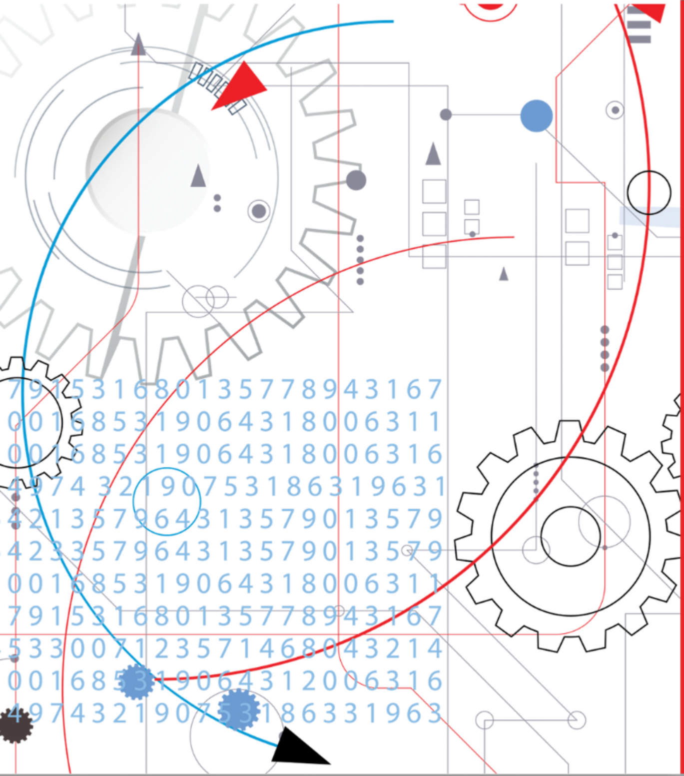


Winning at §112

The Language of Patents

LexisNexis® IP

Speakers

Gene Quinn
IPWatchdog

Jennifer Bailey
Erise IP

David Stitzel
LexisNexis® IP

LexisNexis® IP

AGENDA

Section 112 is where the rubber meets the road with respect to describing an invention. Over the years, however, the level of information that must be present and the specificity and language certainty has only grown. Meanwhile, client budgets have stayed the same.

What a $10,000 budget bought in 1994 is not what it will buy 25-years later, but the Federal Circuit and Patent Office demand more. And now the decision in *Williamson v. Citrix* may just be the tip of the iceberg as Congress contemplates codifying an expansive interpretation of what constitutes functional language as a part of any patent eligibility reform.

There are still tried and true methods to win at 112, and there are tools available to help ensure that 112 issues and inconsistencies that clutter First Office Actions on the Merits won't plague your application.

Today we will discuss all things 112, including what the future may look if 112(f) reform happens, best practices for today, and tools to help you mitigate filing errors before they happen.

Common Problems

Typical §112 rejections and related errors:

- Failing to comply with the written description requirement.
- Claim terms lacking proper antecedent basis.
- Improper dependent or multiple dependent claims.
- Vague and indefinite claim language.
- Functional claiming fatal (or limiting) where the spec fails to adequately disclose (or inadequately discloses) corresponding structure.

Disadvantages of §112 rejections:

- Increased pendency times and unnecessary delays in patent prosecution.
- Increased patent prosecution costs
- Prosecution history estoppel.
- Weaker, less defensible patents.

LexisNexis® IP

112 issues from an Examiner's perspective

Proofread the Claims and Specification. Examiner #1 verified that having to deal with a plethora of formalities before even addressing novelty and inventiveness is, at a minimum, annoying. "I can't stand [section] 112 rejections. Sometimes it's every claim that has an issue."

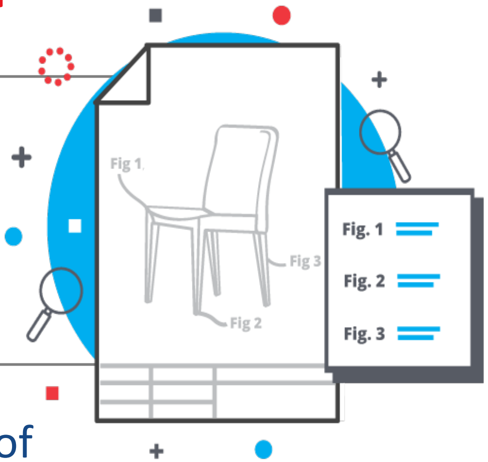
Proofread the Specification and Claims. "When writing up the original disclosure and claims, double check for small errors that can turn into objections or 112 issues," Examiner # 2 said. "Having to guess what the applicants meant can really slow things down, and force applicants into a position where they have to spend more money on examination than they really should have."

Definitions in the Specification. "Adding specific definitions for certain terms in the specification can be very useful when the dictionary provides alternative meanings to what applicants intend. The examiners will use a dictionary definition that allows for the broadest reasonable interpretation because if it can be interpreted a certain way, within reason, the examiner has to determine that nothing else fits that description."

Foreign Translations. "If possible, please check and correct any foreign translations before they get to the examiner. All examiners despise how much time it takes to sort through them when, for most of us, time is our most precious resource."

LexisNexis PatentOptimizer®

Draft higher-quality patent applications built to withstand §112 rejections



Check your claims for proper antecedent basis or possible lack of support in the disclosure, terminology for court rulings, multiple claim sets for new matter, and consistency in part labeling and numbering. Insert pre-formatted citations and automatically renumber claims and references during final editing.

LexisNexis® IP

Nautilus v. Biosig

134 S. Ct. 2120, 2129 (2014

"We conclude that the Federal Circuit's formulation, which tolerates some ambiguous claims but not others, does not satisfy the statute's definiteness requirement. In place of the "insolubly ambiguous" standard, we hold that a patent is invalid for indefiniteness if its claims, read in light of the specification delineating the patent, and the prosecution history, fail to inform, with reasonable certainty, those skilled in the art about the scope of the invention."

"[W]e read §112, ¶2 to require that a patent's claims, viewed in light of the specification and prosecution history, inform those skilled in the art about the scope of the invention with reasonable certainty. The definiteness requirement, so understood, mandates clarity, while recognizing that absolute precision is unattainable. The standard we adopt accords with opinions of this Court stating that 'the certainty which the law requires in patents is not greater than is reasonable, having regard to their subject-matter.'"

Ex parte McAward

PTAB wrote in *Ex parte McAward (Aug. 2017) (PRECEDENTIAL)*:

We recognize that after the Packard decision, the Supreme Court in Nautilus, Inc. v. Biosig Instruments, Inc., 572 U . S . _ , 134 S. Ct. 2120, 2129 (2014), explained that the "definiteness command" of§ 112, i12 "require[s] that a patent's claims, viewed in light of the specification and prosecution history, inform those skilled in the art about the scope of the invention with reasonable certainty." The Court stated that "[t]he definiteness requirement, so understood, mandates clarity, while recognizing that absolute precision is unattainable." Id. We do not understand Nautilus, however, to mandate a change in the Office's approach to indefiniteness…

Thus, in this proceeding, we apply the approach for assessing indefiniteness approved by the Federal Circuit in Packard, i.e., "[a] claim is indefinite when it contains words or phrases whose meaning is unclear." 751 F.3d at 1310, 1314. Put differently, "claims are required to be cast in clear-as opposed to ambiguous, vague, indefinite-terms."

35 U.S.C. 112

(a) In General.—The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor or joint inventor of carrying out the invention.

(b) Conclusion.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or joint inventor regards as the invention.

(c) Form.—A claim may be written in independent or, if the nature of the case admits, in dependent or multiple dependent form.

(d) Reference in Dependent Forms.—Subject to subsection (e), a claim in dependent form shall contain a reference to a claim previously set forth and then specify a further limitation of the subject matter claimed. A claim in dependent form shall be construed to incorporate by reference all the limitations of the claim to which it refers.

(e) Reference in Multiple Dependent Form.—A claim in multiple dependent form shall contain a reference, in the alternative only, to more than one claim previously set forth and then specify a further limitation of the subject matter claimed. A multiple dependent claim shall not serve as a basis for any other multiple dependent claim. A multiple dependent claim shall be construed to incorporate by reference all the limitations of the particular claim in relation to which it is being considered.

(f) Element in Claim for a Combination.—An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

Means Plus Function

A claim limitation not using the phrase "means for" or "step for" triggers a rebuttable presumption that § 112(f) does not apply. **Historically**, the Federal Circuit rarely found a claim limitation to trigger means plus function treatment without the recitation of "means" language in the claim itself. *Apple Inc. v. Motorola, Inc.*, 757 F.3d 1286 (Fed. Cir. 2014).

Means treatment will not apply if persons of ordinary skill in the art reading the specification understand the term used to be the name for the structure that performs the function, "even if the term covers a broad class of structures and even if the term identifies structures by their function. *TecSec, Inc. v. Int'l Bus. Machs. Corp.*, 731 F.3d 1336, 1347 (Fed. Cir. 2013).

Determining whether a claim invokes means treatment cannot merely focus on the presence or absence of the word "means." Focus is placed on "whether the words of the claim are understood by persons of ordinary skill in the art to have a sufficiently definite meaning as the name for structure." *Williamson v. Citrix Online, LLC*, 792 F.3d 1339 (Fed. Cir. 2015).

http://www.ipwatchdog.com/2017/11/15/primer-indefiniteness-means-plus-function/id=89708/

Means Plus Function

The following is a list of non-structural terms that may invoke means plus function treatment: “mechanism for,” “module for,” “device for,” “unit for,” “component for,” “element for,” “member for,” “apparatus for,” “machine for,” or “system for.” This list is not exhaustive, and other non-structural terms may invoke § 112(f).

The following are examples of structural terms that have been found not to invoke means-plus-function treatment: “circuit for,” “detent mechanism,” “digital detector for,” “reciprocating member,” “connector assembly,” “perforation,” “sealingly connected joints,” and “eyeglass hanger member.”

http://www.ipwatchdog.com/2017/11/15/primer-indefiniteness-means-plus-function/id=89708/

Means Plus Function

The Good: Means plus function claiming is an excellent way to make sure that you have captured within the claims all of the various means disclosed in the application. Claiming all the possible permutations can be extremely costly, so using a couple of means-plus-function claims in an attempt to have the claim scope as broad as the disclosure can be a very good strategy.

The Bad: Means plus function claims cannot, however, be the primary claims and don't file patent applications with only means-plus-function claims. Means-plus-function should be used like seasoning — a very powerful garlic! A little may be OK but too much is a very poor choice.

The Ugly: Of course, it is critical to also understand that the rules for means plus function claiming apply very differently with respect to software and computer implemented inventions. *See The Algorithm Cases*.

http://www.ipwatchdog.com/2017/11/15/primer-indefiniteness-means-plus-function/id=89708/

Williamson v. Citrix

792 F.3d 1339 (Fed. Cir. 2015)

- There is still a presumption that when "means" is not used that there is no M+F, but it is no longer a strong presumption and can more easily be overcome.
- To establish no M+F, the limitation must recite to one of ordinary skill in the art sufficiently definite <u>and all the </u>structure <u>required to perform any recited function in the claim</u>.
- Claim limitation at issue in *Williamson*: "… distributed learning control ***module*** for receiving communications ... for relaying the communications ... and for coordinating the operation of the streaming data module."
- CAFC found "module" was a **<u>nonce</u>** term (e.g., generic, black box term) that did not set forth any structure and akin to "means" and, thus, a M+F limitation. Other nonce terms: device, engine, mechanism, member, element, assembly.
- <u>If you seek to avoid use of nonce words</u>, also make sure the name for the limitation connotes sufficiently definite structure to one of ordinary skill (i.e., it is understood as the name for structure).
- *See* MPEP 2181 et seq.

Nonce terms

- Other nonce terms: device, engine, mechanism, member, element, assembly.

- Do not use adjective + nonce term

    - Examples: moving device, pivoting mechanism, light-capturing element
    - **What about:** fiber-optic transmission element, image display device, detent mechanism, reciprocating member?
    - If the adjective for the nonce term is the function claimed, then possibly M+F.

- Make sure the name for the limitation connotes sufficiently definite structure to a skilled person or is understood as the name for structure.

LexisNexis® IP

Software and 112(f)

- To avoid § 112(b) (indefiniteness) don't just describe what is claimed. Describe more sub-steps. Describe an algorithm that performs the claimed function, not just describe the claimed function. **BUT** do describe 100% of the algorithms! Otherwise you will run afoul of 112(f). *Noah Systems v. Intuit,* 675 F.3d 1302 (2012).
- For software, consider avoiding claiming processor, or processor coupled with memory, or computing device, or processor [operable to, configured to, programmed to] perform steps of [ABC]. Instead, describe how the software steps or the structural components interconnect and relate. Use computer-readable medium (CRM) claims and systems claims.
- Describe examples in the specification using language from the claims to tie the structure to the claim terms.
- Include at least some structure (what can you claim that does not narrow because any infringer is going to have to include).
- If receive M+F rejection, traverse it. Often, Examiners are just making a record and do not really believe it is M+F.
- If you can claim a circuit or circuitry, this will usually not be considered M+F.

Means Plus Function

- Means plus function can be useful to get broad-sounding claims, so long as there is a broad (e.g. alternatives and variation) and deep disclosure of the means in the specification. This type of disclosure should be there anyway.

- **Don't go halfway with hybrid claims** like "module for" or "means configured to." Under BRI, **PTO might not apply 112(f) to a hybrid claim where a court will**—worst of both worlds. After *Williamson* weakened the presumption against 112(f) treatment in the absence of "means," it is easier than ever to accidentally get 112(f) treatment. Don't use "means" or "for" unless you want 112(f), especially with nonce words like "mechanism," "element," "device," and "module."

Alice Issues

Failure to disclose underlying structure, materials, or acts in the 112(f) context can also lead to serious problems under *Alice*.

A “wish list” functional claim that was once upon a time popular leads to disaster today. The claims can be both indefinite under 112(b) and (f) and abstract under *Alice* 2A and lack an inventive concept under *Alice* 2B.

No good can come of “wish list” claiming. The description must define a technological solution to a technological problem. The claims must capture the technological solution in definite terms.

Stay away from anything that could even be remotely characterized as functional unless it is done purposefully with eyes wide open. A few functional claims never hurt anyone, and many litigators will tell you they like to have them in a patent.

35 U.S.C. 112(f)

CURRENT LAW:

(f) Element in Claim for a Combination.—An element in a claim for a combination may be **expressed as a means or step** for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

PROPOSED COONS/TILLIS DRAFT:

(f) Functional Claim Elements — An element in a claim **expressed as a specified function** without the recital of structure, material, or acts in support thereof shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

See: https://www.ipwatchdog.com/2019/05/22/draft-text-proposed-new-section-101-reflects-patent-owner-input/id=109498/

LexisNexis® IP

Best Practices for a Robust Disclosure

- Rule against using "the present invention" or "invention" is a reminder of the need to have a thorough and complete specification that describes variations and permutations, not *per se* limiting. *See Absolute Software, Inv. v. Stealth Signal, Inc.*, 659 F. 3d 1121, 1136-1137 (Fed. Cir. 2011). However, if you inartfully describe one piece as "this invention" or "the present invention" you will wind up with that being the invention. *See Honeywell Int'l v. ITT Indus., Inc.*, 452 F.3d 1312 (Fed. Cir. 2006)("The public is entitled to take the patentee at his word and the word was that the invention is a fuel filter.")

- Explaining the functionality of an invention or components is helpful, even necessary to some extent, but function cannot be a substitute for concrete and tangible description (more on this with M+F forthcoming). And ordinarily, how a device is used won't distinguish device claims over prior art anyway. MPEP 2115. And use of "adapted to" or "adapted for" in claims does not necessarily act as a claim limitation. MPEP 2111.04 forthcoming).

- Drawings are your best friend. They force you to expand your written disclosure and whatever one of skill in the art would appreciate is disclosed. File more drawings!

- If your client can't afford for you to draft, file and prosecute all the claims the specification will support why are you afraid of means-plus-function claims? Of course, a little will go a very long way – like a heavy garlic!

Subjective Terminology

Claim scope cannot depend solely on unrestrained, subjective opinion of a particular individual purported to be practicing the invention. *Datamize LLC v. Plumtree Software, Inc.,* 417 F.3d 1342, 1350 (Fed. Cir. 2005). In *Datamize*, the invention was directed to a computer interface screen with an "aesthetically pleasing look and feel." Claims **were held indefinite** because the interface screen may be **"aesthetically pleasing"** to one user but not to another. *Id.* at 1350. See also *Ex parte Anderson,* 21 USPQ2d 1241 (Bd. Pat. App. & Inter. 1991) (the terms "comparable" and "superior" were held to be indefinite in the context of a limitation relating the characteristics of the claimed material to other materials).

In *Interval Licensing LLC v. AOL, Inc.,* 766 F.3d 1364, 1373 (Fed. Cir. 2014), the claim phrase **"unobtrusive manner" was indefinite** because the specification did not "provide a reasonably clear and exclusive definition, leaving the facially subjective claim language without an objective boundary".

During prosecution, **applicant may overcome** a rejection by amending the claim to remove the subjective term, or **by providing evidence that the meaning can be ascertained by one of ordinary skill in the art after reading the disclosure**. *See DDR Holdings, LLC v. Hotels.com, L.P.,* 773 F.3d 1245, 1260 (Fed. Cir. 2014)(terms of degree, specific and unequivocal examples may be sufficient to provide a skilled artisan with clear notice of what is claimed.*)*

Subjective Terminology

“For some facially subjective terms, the definiteness requirement is **not satisfied by merely offering examples** that satisfy the term within the specification.” *DDR Holdings, LLC v. Hotels.com, L.P.,* 773 F.3d 1245, 1260 (Fed. Cir. 2014).

In *DDR* attempts to characterize **"look and feel"** as purely subjective failed. The specification defined “look and feel” in several locations. The examples provided were consistent with established meaning of the term “look and feel” in the art. The Federal Circuit concluded: “In sum, **‘look and feel’ is not a facially subjective term** like ‘unobtrusive manner’ in *Interval* or "aesthetically pleasing" in *Datamize.”*

Tinnus Enterprises v. Telebrands

846 F.3d 1190, 1206 (Fed. Cir. 2017)

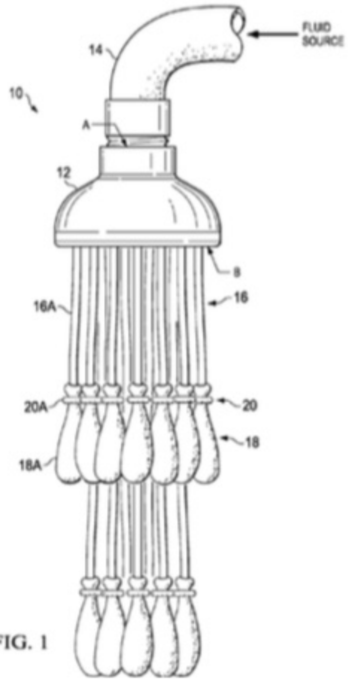


FIG. 1

Telebrands contended that the claim term "substantially filled" is indefinite because a POSA would not be able to determine whether an expandable container meets this limitation.

The Federal Circuit wrote:

“Telebrands asserts that a POSA has "a general knowledge about and experience with expandable containers, including without limitation balloons, and at least an associate's degree in science or engineering." J.A. 835-36 ¶¶ 10-13. We find it difficult to believe that a person with an associate's degree in a science or engineering discipline who had read the specification and relevant prosecution history would be unable to determine with reasonable certainty when a water balloon is *substantially filled*.”

LexisNexis® IP

Negative Limitations

Applicants may use negative limitations, or any style of expression or format of claim which makes clear the boundaries of the subject matter for which protection is sought. *In re Swinehart,* 439 F.2d 210 (CCPA 1971).

There is nothing inherently ambiguous or uncertain about a negative limitation. So long as the boundaries of the patent protection sought are set forth definitely, albeit negatively.

”Negative claim limitations are adequately supported when the specification describes a reason to exclude the relevant limitation." *Santarus, Inc. v. Par Pharm.,* 694 F.3d at 1344, 1351 (Fed. Cir. 2012) ("wherein the composition contains no sucralfate").

Properly describing alternative features—without articulating advantages or disadvantages of each feature—can constitute a "reason to exclude.” *Inphi Corp. v. Netlist, Inc.*, 805 F.3d 1350 (Fed. Cir. 2015)("the chip selects of the first and second number of chip selects are DDR chip **selects *that are not CAS, RAS, or bank address signals***”) (emphasis added).

There is no heightened description requirement in order to use negative limitations. Using the customary written description requirement, the question is whether persons of ordinary skill in the art recognize what has been invented. *Nike, Inc. v. Adidas AG*, 812 F.3d 1236 (Fed. Cir. 2016).

LexisNexis® IP

U.S. Patent No. 8,245,301 (issued Aug. 14, 2012)

1. A network monitoring and visualization system comprising… such that the intelligent icons displayed on the display device represent **only** the most recent data values of the corresponding MDL model.

Argument from *Amendment* filed February 28, 2012:

“Moreover, Foslien describes generating essentially a timeline display that includes previous values of variables. This is in contrast to Applicants’ invention in which intelligent icons represent only the most recent data values of a corresponding MDL model.

Because Applicant’s invention is concerned with generating a display that helps an operator visualize complex underlying data, not including historical values can help reduce clutter on a display and can help prevent information overload.”

Numerical Ranges and Amount Limitations

Open-ended numerical ranges should be carefully analyzed for definiteness. For example, when an independent claim recites a composition comprising “at least 20% sodium” and a dependent claim sets forth specific amounts of non-sodium ingredients which add up to 100%, apparently to the exclusion of sodium, an ambiguity is created with regard to the “at least” limitation.

The term “up to” includes zero as a lower limit, *In re Mochel,* 470 F.2d 638 (CCPA 1974); and “a moisture content of not more than 70% by weight” reads on dry material, *Ex parte Khusid,* 174 USPQ 59 (Bd. App. 1971).

The phrase “an effective amount” may or may not be indefinite, depending upon whether one skilled in the art could determine specific values for the amount based on the disclosure.

http://www.ipwatchdog.com/2016/01/30/patent-drafting-relative-terminology-can-be-dangerous/id=65455/

Meaning of “About”

“About" is given its ordinary meaning of "approximately" unless clearly redefined in the specification. *See Merck & Co., Inc. v. Teva Pharms. USA, Inc.,* 395 F.3d 1364, 1369-70 (Fed. Cir. 2005).

In *Ex parte Oetiker,* 23 USPQ2d 1641 (Bd. Pat. App. & Inter. 1992), the phrases “relatively shallow,” “of the order of,” “the order of about 5mm,” were indefinite because the specification lacked some standard for measuring the degrees intended.

Determining range encompassed by the term "about” requires consideration of context as used in the specification. *Ortho-McNeil Pharm. v. Caraco Pharm. Labs.,* 476 F.3d 1321 (Fed. Cir. 2007).

In *W.L. Gore & Associates, Inc. v. Garlock, Inc.,* 721 F.2d 1540 (Fed. Cir. 1983), rate “exceeding about 10% per second” was definite because infringement could clearly be assessed through the use of a stopwatch.

In ” *Amgen, Inc. v. Chugai Pharmaceutical Co.,* 927 F.2d 1200 (Fed. Cir. 1991), “at least about” was indefinite because there was close prior art and there was nothing in the specification, prosecution history, or the prior art to provide any indication as to what range of specific activity is covered by the term “about.

LexisNexis® IP

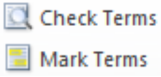
Business Method Profanity

price, cost, bid, auction, financial, bank, account, buy, sell, commerce, insurance, negotiation, trade, stock, ticker, exchange, money, business, bill, invoice, discount, coupon, and advertise

LexisNexis® IP

LexisNexis PatentOptimizer®

Check Terms / Mark Terms

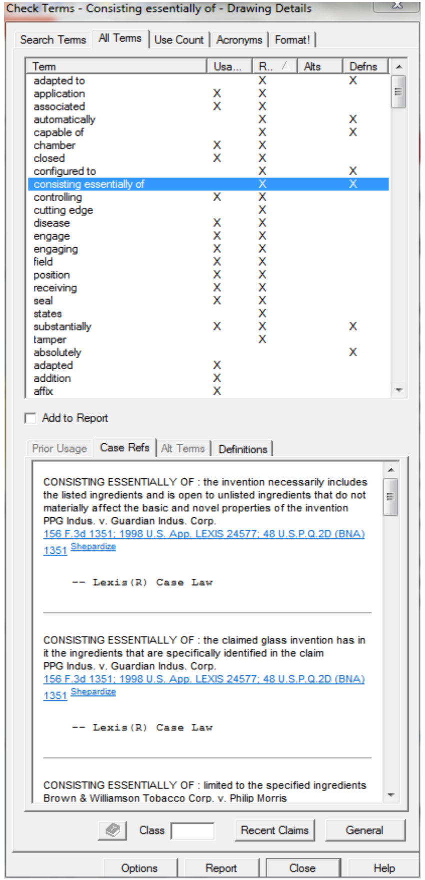


- **Prior Usage** – Identifies claim terms/phrases with Prior Usage in the claims of other patents, as referenced in our **Matthew Bender® Attorney's Dictionary of Patent Claims (ADPC)** treatise.

- **Case References** – Complimentary bundled access to patent cases where the court construed the meaning of the claim term/phrase at issue, as well as claim term/phrase meanings presented by the litigants, from **LexisNexis federal case law (US Supreme Court, CAFC, District Court decisions)** and **administrative rulings (PTAB, BPAI)**.

- **Alternative Terms** – Identifies broader, narrower, interchangeable, synonymous, related, and alternative terms/phrases from our **Proprietary LexisNexis Patent Thesaurus, Matthew Bender® Attorney's Dictionary of Patent Claims (ADPC)** treatise, and **Elsevier's Compendex® Engineering Thesaurus**.

- **Definitions** – Identifies potentially problematic terms/phrases that could be construed as being vague and indefinite, functional, impacting claim breadth and scope, invoking public relations concerns, or considered offensive, based on a series of **Proprietary Patent Profanity Dictionaries**.



LexisNexis® IP

LexisNexis PatentOptimizer®

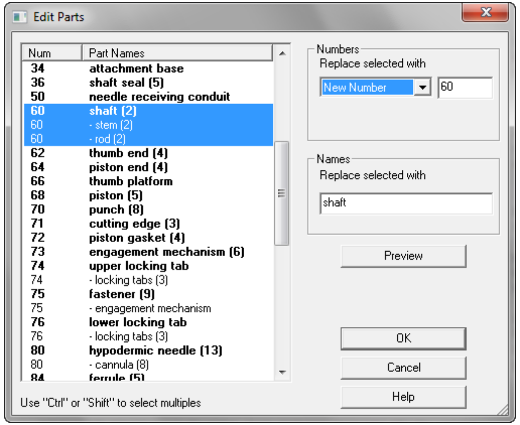
Check Parts

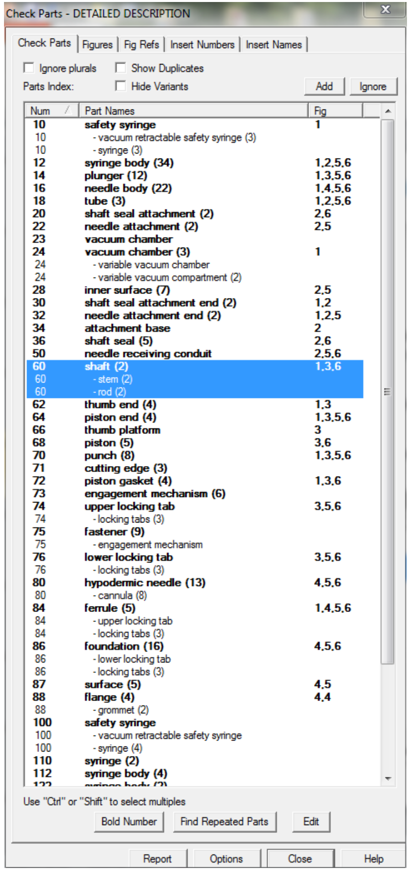
Check Parts

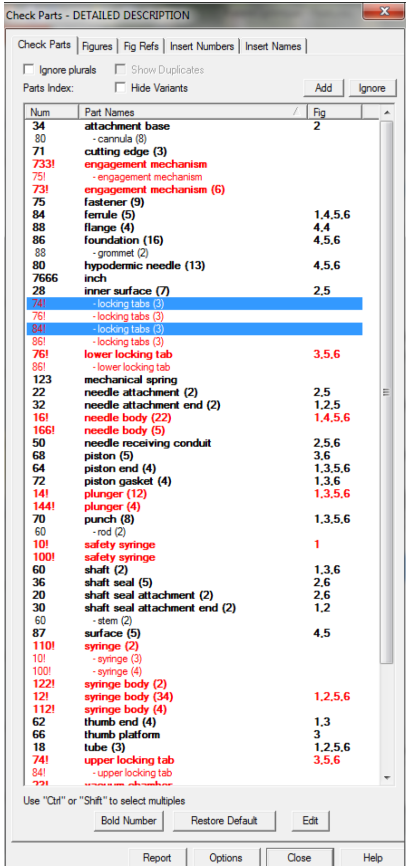
• Checks the **accuracy** and **consistency** of part names/numbers within the drawings and specification.

• **Same** part number identifying **different** part names.

• **Different** part numbers identifying **same** part name.





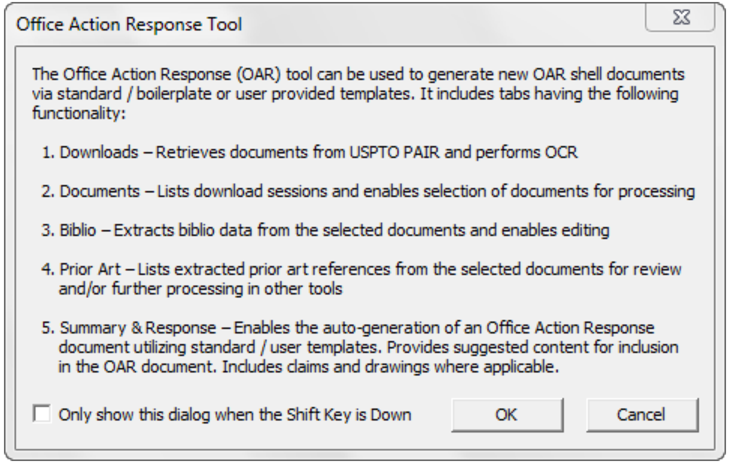


LexisNexis® IP

LexisNexis PatentOptimizer®

OA Response

- **Improve patent prosecution efficiency**.
- **Streamline your analysis** of Office Action (OA) rejections, cited prior art references, objections, and indications of allowable subject matter.
- Auto-generate a **comprehensive OA Response shell** containing desired substantive and customizable boilerplate response language.
- **Improve consistency** of OA Responses that are custom-tailored for a specific client and/or patent attorney.
- **Improve accuracy** of OA Responses and **facilitate avoidance** of Non-Compliant Amendments and Non-Responsive Responses.



LexisNexis® IP

For more information, contact:

Morgan Connell

mconnell@lexisnexisip.com
215-441-6466

https://www.lexisnexisip.com/patent-optimizer/

LexisNexis® IP